Student number:	Seat No:

Mid-Semester Test – Sm1 2026

FNCE30007
Writing Time: 60 minutes
Reading Time: None
Weighting: 25%

Permitted Materials

- Calculator: Casio FX82 (any suffix)
- Mathematical instruments (Protractor, compass, ruler, circle template).
- Formula sheet provided.

Instructions to Students

- During writing time, clearly write your student number and seat number on the cover sheet in the above boxes only. Ensure each digit is clearly written.
- Please write the answers in CAPITAL LETTERS only in the designated answer boxes provided within the exam paper for each question.
- Use a pen and not a pencil to write your answers
- Do not communicate with other students in the exam venue.
- No part of the exam paper may be removed from the venue.
- Ensure you only have standard permitted materials on your desk plus additional materials if listed above.
- Raise your hand if you require assistance.
- Students are NOT permitted to take pictures or make any reproduction via any means of any of the questions in the exam paper.

- Test contains 13 multiple-choice questions.
- Total marks: 15.
- Questions 5 and 6 are worth 2 marks each. All other questions are worth 1 mark.
- Select one answer to each question.

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Question 1 (1 mark)

A wheat producer hedges by taking a short position in wheat futures and needs to close out the hedge before the expiration of the futures contracts. Just before the hedge is closed out, however, the basis (spot - futures) unexpectedly strengthens. Relative to what had been expected prior to the strengthening of the basis, which of the following is correct?

A. The producer is now speculating rather than hedging.
B. One cannot tell unless the initial margin is known.
C. The effective price received by the producer is lower than expected.
D. The effective price received by the producer is higher than expected.
E. The effective price received is unchanged because futures eliminate basis risk.

Write your CAPITAL LETTER answer in the box

Question 2 (1 mark)

In the Metallgesellschaft discussion, why could a stack-and-roll hedge create significant problems for the hedger even if it made economic sense over the long run?

A. Because exchange-traded futures cannot be closed out before maturity.
B. Because short-dated futures eliminate all basis risk, leaving only counterparty risk.
C. Because daily margin calls and repeated rollovers can create severe short-run liquidity pressure before gains on the long-dated exposure are realized.
D. Because forwards, not futures, must be used whenever the exposure is longer-dated than one year.
E. Because a long futures hedge necessarily loses money whenever spot prices fall.

Write your CAPITAL LETTER answer in the box

Question 3 (1 mark)

One of the articles discussed in class noted that commercial long cocoa futures positions by chocolate makers fell sharply. Which explanation best matches the class discussion?

A. Chocolate makers are naturally long cocoa, so reducing long futures lowers their downside exposure.
B. Chocolate makers are naturally short cocoa, and reducing long futures removes all price risk.
C. Chocolate makers are naturally long cocoa, and OTC insurance is always cheaper than exchange-traded futures.
D. The fall in commercial long positions proves cocoa had become a pure investment asset.
E. Chocolate makers are naturally short cocoa, but higher cocoa prices increased the cash needed to finance margins and some firms preferred short-term buying while waiting for prices to fall.

Write your CAPITAL LETTER answer in the box

Question 4 (1 mark)

During class we discussed supply risk due to the situation in the Strait of Hormuz. If prompt Brent rises much more than deferred Brent because physical barrels become more valuable to hold now, which statement is most accurate?

A. The market is moving toward deeper backwardation and a higher implied convenience yield.
B. The change proves futures prices must now exceed forward prices.
C. The basis at every maturity must be zero.
D. The risk-free rate must have fallen below zero.
E. The market is moving toward stronger contango and a lower implied convenience yield.

Write your CAPITAL LETTER answer in the box

Question 5 (2 marks)

A diversified equity portfolio is worth $28,000,000 and currently has a beta of 1.25. The portfolio manager wants to reduce the beta to 0.65 for the next five months using S&P 500 futures. The futures contract the manager wants to rely on is trading at 4,650, and each contract is $50 times the index. Ignoring any small maturity mismatch, which is the most appropriate futures trade?

A. Long 96 contracts
B. Short 72 contracts
C. Short 60 contracts
D. Long 72 contracts
E. Short 96 contracts

Write your CAPITAL LETTER answer in the box

Question 6 (2 marks)

A 9-month futures contract is written on silver, an investment commodity. The spot price is $25.00 per ounce. The storage cost over the next 9 months is $0.40 per ounce, payable at the end of the 9 months. The risk-free rate is 4% per annum with continuous compounding. The 9-month futures price is quoted at $27.00 per ounce. Ignoring transaction costs, which statement is correct?

A. There is no arbitrage because silver may have a convenience yield just like a consumption commodity.
B. Go long futures and short silver, earning about $0.84 per ounce.
C. Go short futures and buy silver spot financed by borrowing, earning about $1.24 per ounce.
D. Go short futures and buy silver spot financed by borrowing, earning about $0.84 per ounce.
E. There is no arbitrage because positive storage costs allow the futures price to exceed the spot price by any amount.

Write your CAPITAL LETTER answer in the box

Question 7 (1 mark)

A trader goes long 5 crude-oil futures contracts at $72.40 per barrel. Each contract is on 1,000 barrels. The initial margin is $4,800 per contract and the maintenance margin is $3,600 per contract. At the end of the first trading day, the settlement price is $71.10 per barrel. Which statement is correct?

A. No margin call is required; the margin-account balance is $17,500.

B. A margin call of $500 is required, because the account only needs to be restored to the maintenance margin.

C. A margin call of $6,500 is required, because the balance must be restored to the initial margin.

D. A margin call of $6,000 is required, because the futures price fell by $1.30 per barrel.

E. A margin call of $5,500 is required, because the loss is measured relative to the maintenance margin.

Write your CAPITAL LETTER answer in the box

Question 8 (1 mark)

Which pair of statements best reflects the lecture discussion of pros and cons of hedging?

A. In favor: hedging is useful only for firms with poorly diversified shareholders. Against: a hedge fails whenever it loses money.

B. In favor: managers should speculate on commodity and FX prices because they know their industry. Against: shareholders are never diversified.

C. In favor: hedging always lowers risk even if competitors do not hedge. Against: gains on the underlying can never offset hedge losses.

D. In favor: hedging guarantees higher expected profits. Against: hedging eliminates quantity risk.

E. In favor: firms should reduce non-core market risks and focus on their main business. Against: diversified shareholders can often make their own hedging decisions.

Write your CAPITAL LETTER answer in the box

Question 9 (1 mark)

The current spot price of copper is $100. The 6-month futures price is $104. Analysts expect the spot price in 6 months to be $102. Which statement is correct?

A. The market is in "industry" backwardation and normal contango.
B. The market is in "industry" contango and normal contango.
C. One cannot classify the market without knowing storage costs.
D. The market is in "industry" backwardation and normal backwardation.
E. The market is in "industry" contango and normal backwardation.

Write your CAPITAL LETTER answer in the box

Question 10 (1 mark)

A New Zealand importer must pay USD 4 million in three months. To lock in the NZD cost using the forward market, the importer should:

A. go long USD forwards because it is naturally short USD.
B. short USD spot today and invest the proceeds.
C. go short USD forwards because it is naturally short USD.
D. go long NZD forwards because it is naturally long USD.
E. do nothing because forwards can hedge only known foreign-currency inflows.

Write your CAPITAL LETTER answer in the box

Question 11 (1 mark)

A European call option on a stock has current stock price $52, strike price $50, and premium $4.80. Which statement is correct?

A. Intrinsic value = $4.80; time value = $0.00
B. Intrinsic value = $0.00; time value = $4.80
C. Intrinsic value = $2.80; time value = $2.00
D. Intrinsic value = $2.00; time value = $2.80
E. Intrinsic value = $50.00; time value is negative

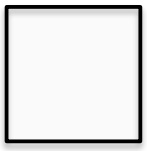
Write your CAPITAL LETTER answer in the box

Question 12 (1 mark)

Which statement is correct for a European call on a non-dividend-paying stock?

A. Its value must always be at least equal to the strike price.
B. Its lower bound is exactly zero for every strike and maturity.
C. Its value cannot exceed the current stock price.
D. It is always worth more than the corresponding American call.
E. Its value can exceed the current stock price if it is in the money.

Write your CAPITAL LETTER answer in the box



Question 13 (1 mark)

A Reuters article discussed in class noted that the Middle East war had made currency hedging more expensive for Indian importers with USD payables. Which statement best explains that point?

A. Rupee call options became cheaper than dollar call options.
B. Hedging currency risk was no longer feasible once oil prices became a major macroeconomic risk.
C. Higher oil prices necessarily make the domestic risk-free rate fall, which is why every currency forward becomes more expensive.
D. Importers with dollar payables are naturally long USD, so they must short USD forwards when hedging.
E. Rising forward premiums made forward hedges more expensive, while higher volatility made protective option hedges more expensive.

Write your CAPITAL LETTER answer in the box

